

Question No:6

Question description

There are n boys and m girls in a school. Next week a school dance will be organized. A dancing pair consists of a boy and a girl, and there are k potential pairs.

Your task is to find out the maximum number of dance pairs and show how this number can be achieved.

Constraints

- $1 \leq n, m \leq 500$
- $1 \leq k \leq 1000$
- $1 \leq a \leq n$
- $1 \leq b \leq m$

Input

The first input line has three integers n, m and k: the number of boys, girls, and potential pairs. The boys are numbered $1, 2, \dots, n$, and the girls are numbered $1, 2, \dots, m$.

After this, there are k lines describing the potential pairs. Each line has two integers a and b: boy a and girl b are willing to dance together.

Output

First print one integer rr: the maximum number of dance pairs. After this, print lines describing the pairs. You can print any valid solution.

Logical Test Cases

Test Case 1

INPUT (STDIN)

3 2 4
1 1
1 2
2 1
3 1

EXPECTED OUTPUT

2
1 2
2 1

Test Case 2

INPUT (STDIN)

3 2 4
1 1
1 2
2 3
3 2

EXPECTED OUTPUT

3
1 1
2 3
3 2

Code:

```
1  #include <stdio.h>
2  #include <string.h>
3
4  #define MAX 505
5
6  int n, m, k;
7
8  /* Adjacency matrix:
9   |   graph[boy][girl] = 1 means they can dance together
10 */
11 int graph[MAX][MAX];
12
13 /* matchGirl[g] = boy currently matched with girl g */
14 int matchGirl[MAX];
15
16 /* Visited girls during one DFS */
17 int visited[MAX];
18
19 /*
20 |   Try to find a girl for boy u.
21 */
22 int bpm(int u)
23 {
24     for (int v = 1; v <= m; v++)
25     {
26         if (graph[u][v] && !visited[v])
27         {
28             visited[v] = 1;
```

```

29
30     /*
31     |     If girl v is free OR
32     |     the boy currently matched with v
33     |     can be moved to another girl
34     | */
35     if (matchGirl[v] == 0 || bpm(matchGirl[v]))
36     {
37         matchGirl[v] = u;
38         return 1;
39     }
40     }
41 }
42
43 return 0;
44 }
45
46 int main()
47 {
48     scanf("%d %d %d", &n, &m, &k);
49
50     /* Read all possible pairs */
51     for (int i = 0; i < k; i++)
52     {
53         int boy, girl;
54
55         scanf("%d %d", &boy, &girl);
56
57         graph[boy][girl] = 1;
58     }
59
60     /* Initially no girl is matched */
61     for (int i = 1; i <= m; i++)
62         matchGirl[i] = 0;
63
64     int maximumPairs = 0;
65
66     /*
67     |     Try to match every boy.
68     | */
69     for (int boy = 1; boy <= n; boy++)
70     {
71         memset(visited, 0, sizeof(visited));
72
73         if (bpm(boy))
74             maximumPairs++;
75     }
76

```

```
76
77     printf("%d\n", maximumPairs);
78
79     /*
80      matchGirl[g] = b
81      means boy b dances with girl g.
82     */
83     for (int girl = 1; girl <= m; girl++)
84     {
85         if (matchGirl[girl] != 0)
86         {
87             printf("%d %d\n", matchGirl[girl], girl);
88         }
89     }
90
91     return 0;
92 }
```

Output:

Test Case-1:

```
3 2 4
1 1
1 2
2 1
3 1
2
2 1
1 2
PS C:\Users\kesha\OneDrive
```

Test Case-2:

```
3 2 4
1 1
1 2
2 3
3 2
2
1 1
3 2
```

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